

Interview Protocol

Supplemental material for “Debugging as Evidence-Driven Reasoning: Visualization Opportunities in Data-Intensive Programming” (IEEE VIS 2026 Short Papers).

Title: Understanding User Needs and Opportunities for Supporting Debugging with Visualization Tools

Study type: Semi-structured interview (Phase 1 of a mixed-method study)

Duration: ~30–45 minutes

Recording: Audio only (with consent)

1. Introduction (~3–4 min)

Who I am.

Hi! Thanks so much for meeting with me today. My name is Yongbo Chen, and I’m a Ph.D. student supervised by Prof. Rebecca Faust at Tulane University. My research focuses on debugging and visualization tools, especially for data-intensive programming.

Why we’re here.

The goal of this interview is to better understand the real-world challenges and workflows people experience when debugging data-intensive code. In particular, we are interested in what kinds of visualization tools or intelligent assistance — such as large language models — could help make debugging easier, faster, or less frustrating. We hope to learn what specific information, features, or ways of presenting data you would find most useful when trying to understand how your code behaves or why something is not working as expected. Your responses will help us design more helpful visualization or intelligent interfaces (such as natural language support).

Confidentiality.

Before we begin, I’d like to assure you that everything you share will be kept confidential. Your identity will not be revealed in any reports or papers. The session should take around 30 to 45 minutes.

Consent.

[Participant reads the consent script.]

Is it okay if I record our conversation, for analysis purposes only?

[Wait for verbal consent.]

Great, I’ll go ahead and start recording.

[START RECORDING]

Just to confirm again — do you consent to participate and to be recorded? [Wait for verbal consent.]

What we will do.

As mentioned, the goal of this study is to understand the real-world challenges and workflows people experience when debugging data-intensive code. I'll ask you a series of questions about your typical debugging process, the tools you use, and what would make debugging easier or more intuitive for you. To respect your time, if we spend too long on one topic, I may gently move us along so we stay within the planned 30–45 minutes. Let's get started.

2. Warm-Up & Background (~5 min)

Your background & work - What is your current area of study or work?

Programming & debugging experience - What programming languages or tools do you use most frequently? - How would you describe your comfort level with debugging (e.g., beginner, intermediate, advanced)? - Do you mostly debug to fix code that won't run, or more to check and validate outputs against your domain knowledge?

3. Main Interview Questions (~20–25 min)

A quick note before we start: When we say “debugging,” we don't just mean fixing bugs inside the program. We also include times when you're trying to understand what your code is doing — such as checking variable values, making sense of unexpected results, or verifying whether code output aligns with policies, regulations, or common sense. Even if you don't write much code or rarely chase down technical bugs, your process of reasoning about outputs, spotting inconsistencies, or questioning unexpected results is still part of debugging for this study. So feel free to talk about any experience where you were trying to understand how code works or why it behaves a certain way, whether or not it involved a specific bug.

Question Set 1 — Debugging Experience (~7–8 min) - Can you describe a recent debugging task you had to deal with? For example, when you noticed a bug or unexpected behavior, tried to figure out why it happened, and worked to fix it. - What were you trying to understand or fix? - What specific steps or tools did you use? - Was anything particularly frustrating or confusing during that process? For example, difficulty finding where a bug starts, understanding variable values, or figuring out code flow. - Are there any other pain points or challenges you've encountered during debugging that we haven't discussed?

Question Set 2 — Information-Seeking (~6–7 min) - What information do you usually look for when debugging? (e.g., variable values, execution order, function behavior) - Are there times when you can't find what you need? Why? - What caused that frustration? For example, are you looking for variable values, function call sequences, performance metrics, or something else? - What strategies do you use when you're stuck?

Question Set 3 — Tool Evaluation (~6–7 min) - What tools do you typically use to debug? This could include print statements, IDE debuggers,

visualization tools, or AI assistants. - If you are an AI-heavy user, why do you prefer AI? - What do you like or dislike about those tools? - Do you use any visual tools or interactive views (e.g., tracing diagrams, call graphs)? - Do you find them helpful? Why or why not?

4. Forward-Looking Reflection on Future Tool Support (~7–8 min)

We'd like to hear your thoughts on future tools, including AI-based ones. - First: If you could have any kind of support tool, what would you wish for? (e.g., better visualizations, smarter hints, automated explanations) - Then: Recently there's been growing interest in using large language models — like ChatGPT — to support debugging. Have you ever used, or thought about using, tools like ChatGPT or Copilot to help with debugging? - Main question: What kinds of information or visualizations would you want such a system to automatically provide? For example, summaries of variable changes, call graphs, or anomaly highlights? - How would you prefer these to be presented — inline while coding, in a separate panel, or as a report afterward?

5. Wrap-Up (~3–5 min)

- We've reached the end of our interview.
- Was there anything you wanted to share but didn't get the chance to?
- Do you have any feedback about this interview?
- Would you be open to us following up with a short survey later on?
- Do you know anyone else who might be a good fit for this study?
- [Note-taker, if present: any questions you'd like to ask?]
- Do you have any questions before we end the meeting?

Thank you! We really appreciate you taking the time to speak with us today, and thank you for your participation and invaluable insights. Your answers will be recorded and summarized anonymously for research and publication. If you have any concerns, please let us know.